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# Indoor positioning system for single LED light based on deep residual shrinkage network

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ABSTRACT

To enhance the indoor visible light positioning (VLP) system's performance, this paper proposes a deep residual shrinkage network (DRSN)-based VLP system utilizing a single LED and multiple photodetectors. This neural network effectively resolves issues prevalent in the training phase of conventional neural networks, such as vanishing or exploding gradients, noise interference, and indistinct feature characterization, by incorporating residual learning, soft threshold functions, and an attention mechanism. This approach contributes to a more stable and efficient training process. Simulation results illustrate that the DRSN algorithm produces an average localization error of 1.02 cm, with a maximum error of 4.35 cm in a 3.6 m × 3.6 m × 3 m space. During the experimental phase, the average localization error remains below 6 cm, with 90% of the errors below 10 cm. Comparative analysis with alternative localization algorithms reveals the proposed method's ability to achieve precise localization accuracy.

## 1. Introduction

With the continuous innovation and technological advancements, positioning-based services have gained increasing popularity among users. While the global positioning system (GPS) effectively caters to daily life needs, it encounters challenges, especially in complex indoor environments, due to susceptibility to multipath fading, leading to significant positioning errors [1]. Consequently, researchers have explored numerous indoor positioning methods, including ultrasonic, Wi-Fi, radio frequency identification, ZigBee Bluetooth, and ultra-wideband infrared, among others [2–6]. However, these approaches commonly exhibit high susceptibility to electromagnetic radiation interference, high deployment costs, and reduced positioning accuracy [7]. Conversely, visible light positioning (VLP) has emerged as a research focal point in the wireless localization field [8,9], offering advantages such as abundant bandwidth resources, cost-effectiveness, and immunity to electromagnetic interference.

Currently, the more traditional localization methods encompass angle of arrival (AOA), time of arrival (TOA), time difference of arrival (TDOA), and received signal strength (RSS) [10–13]. Scholars have integrated these methods to enhance localization performance. Yang et al. introduced a 3D visible light indoor positioning system utilizing AOA and RSS, employing an LED array and tilted receivers to eliminate inter-cell interference. Experimental results demonstrate an average error distance of less than 3 cm [14]. In fingerprint localization, the objective is to establish the mapping relationship between RSS values and corresponding coordinates. Researchers leverage machine learning and deep learning algorithms [15], including convolutional neural networks (CNN), long short-term memory networks (LSTM), and gated recurrent unit networks (GRU) [16–18]. Concurrently, researchers combine various machine learning algorithms to further enhance localization accuracy. Hsu et al. proposed an RSS-based visible light positioning (VLP) system using CNN and data preprocessing to address under-illuminated regions. Results indicate that, at a 200 cm distance between the LED transmitter and receiver, the average localization error using only the CNN model is 10.19 cm, while the CNN with RSS preprocessing achieves an average localization error of 5.31 cm [19]. Wu et al. presented an RSS-based VLP system using sigmoid function data preprocessing (SFDP) method, applied to two regression-based machine learning algorithms: the second-order linear regression machine learning (LRML) algorithm and the kernel ridge regression machine learning (KRRML) algorithm. Experimental results show significantly improved positioning accuracy for LRML and KRRML algorithms using the SFDP method [20]. Chen et al. proposed an indoor high-precision 3D positioning system based on the improved immune particle swarm optimization (IIMPSO) algorithm. They optimized the field-of-view

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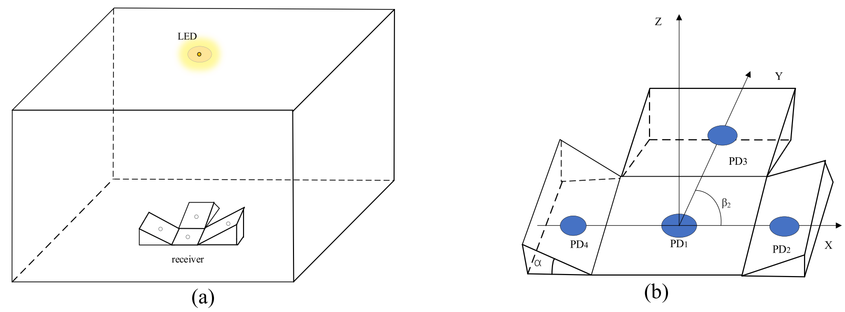



**Fig. 1.** system model (a) Visible light localization system model (b) Receiver model.

angle of the indoor receiver to minimize the effect of reflection selection and employed the kalman filtering algorithm to reduce external environmental influence. Finally, the IIMPSO algorithm optimized the search in space, improving positioning accuracy and convergence speed [21]. However, these schemes often use three or even nine LEDs, rendering them impractical in scenarios with sparse LED fixtures.

Some researchers have used a smaller number of LEDs to achieve the same high precision localization [22,23]. In the context of indoor visible light positioning, Hao et al. focused on sparse LED deployment, utilizing a single LED fixture. They achieved a remarkable indoor localization accuracy of 90% within 0.16 m by employing a smartphone as the receiver [24]. Liu et al. developed an indoor visible light localization system utilizing a single LED and a rotatable photodetector (PD). Their approach segmented the entire room into internal and four external zones using a random forest (RF) algorithm. The interior region was directly localized with a rotatable PD, while the exterior region leveraged an extreme learning machine (ELM) and density-based spatial clustering of applications with noise (DBSCAN) to enhance accuracy near interior walls and corners. Simulation results indicated a notable reduction in the average localization error of the exterior region to 1.43 cm with the proposed rotary localizer and hybrid algorithm, resulting in an overall room accuracy of 1.74 cm [25]. Cheng et al. proposed a visible light positioning system based on individual LED communication and an optical camera. This system utilized geometrical features of LED projection and the intersection line on a flat surface to address receiver tilt, achieving high localization accuracy for tilt angles ranging from $\pm 45^{\circ}$ to $\pm 40^{\circ}$ [26].

Moreover, researchers have empirically demonstrated that strategically adjusting the tilt angle of the receiver can substantially augment the channel capacity, thereby enhancing the visible light positioning (VLP) system's overall performance [27]. Yu et al. proposed a visible light positioning system employing a single LED and a novel receiver configuration. This receiver comprises a horizontal photodetector and two tilted PDs, with a detailed analysis of the impact of tilt and azimuth angles on positioning error. Experimental findings indicate that optimal accuracy is achieved with a tilt angle of $20^{\circ}$ and an azimuth angle within the range of $30^{\circ}$–$90^{\circ}$ [28]. Building upon this foundation, Qin et al. utilized three tilted photodetectors and one horizontal PD in their receiver design. Their investigation delved into the effects of PD tilt angles, azimuth angles, and distances between horizontal and tilted PDs on localization accuracy. Simulation results reveal that, within a 1 m $\times$ 1 m $\times$ 1.5 m positioning area, the best accuracy is attained with a tilt angle ($\alpha$) of $25^{\circ}$, azimuthal difference ($\theta$) of $90^{\circ}$, placement angles ($\omega1$, $\omega2$, $\omega3$) of $0^{\circ}$, $90^{\circ}$, and $180^{\circ}$ respectively, and a 1 cm distance between the center of the horizontal PD and the center of the tilted PDs [29]. Han et al. employed a receiver comprising one horizontal PD and four tilted PDs. Simulation outcomes demonstrate high localization accuracy when the tilt angle of the PDs does not exceed $30^{\circ}$ [30]. Importantly, this structured receiver type offers a distinct advantage by effectively diversifying data features, facilitating enhanced learning and understanding of signal changes in the environment by the network. This comprehensive capture of signal characteristics from different receivers enhances the network's sensitivity to various locations, leading to significantly improved localization performance.

Hence, we have constructed a visible light positioning system utilizing a single LED and multiple tilted photodetectors. Additionally, we propose a novel visible light positioning algorithm based on a deep residual shrinkage network (DRSN). This network enhances both performance and efficiency through the incorporation of a residual structure, facilitating multiple levels of abstract representation. The integration of a soft threshold function not only aids the network in processing anomalous and noisy data but also diminishes redundant information within the model, enhancing the network's adaptability to disturbances or noises. Concurrently, the introduction of squeeze and excitation empowers the network to allocate more attention to crucial channel features, thereby elevating the characterization ability of these features.

We emphasize the following contributions:

1. Our investigation unveils the utilization of fingerprint identification and deep residual shrinkage network algorithms in multiple photodetector-based VLP systems for precise localization, marking the pioneering nature of this work.
2. We perform distinct examinations to assess the localization accuracy of the deep residual shrinkage network under varying heights and different datasets. This detailed scrutiny serves to validate and strengthen the algorithm's generalization capabilities.
3. We conduct a comparative analysis of the DRSN against prevalent algorithms. Our experimental findings indicate the superior performance of DRSN over the other three algorithms within the specified settings.

The rest of the paper is organized as follows: first in Section 2, we describe the components of the indoor VLP localization system model. Then in Section 3 the localization principles are described. Then the simulation and experimental results are analyzed in Section 4. Finally in Section 5, conclusions are provided for this study.

## 2. Visible light positioning model and system design

### 2.1. System model

The experimental area measures 3.6 m $\times$ 3.6 m $\times$ 3 m, an LED, positioned at coordinates (1.8, 1.8, 3), serves as the illumination and signal transmitter atop the model, and the receiver placed at the bottom of the model consists of four photodetectors PDs, with the centermost one placed horizontally (denoted as PD1), and the remaining three placed according to a certain angle of inclination (denoted as $PD_i$, $i = 2$, 3, 4). Fig. 1 displays the models of the visible light localization system and the receiver.

The positional relationship between the horizontal photodetectors and each tilted photodetector can be expressed by equation (1):

$$\begin{cases} X_{li} = X_l + l\cos(\alpha)\cos(\beta_i) \\ Y_{li} = Y_l + l\cos(\alpha)\sin(\beta_i) \\ Z_{li} = Z_l + l\sin(\alpha) \end{cases} \tag{1}$$

where $(X_{li}, Y_{li}, Z_{li})$ and $(X_l, Y_l, Z_l)$ are the coordinates of the tilt photodetector and the horizontal photodetector, respectively, $l$ is the distance between the center of the tilt photodetector and the center of the horizontal photodetector, and $\alpha$ is the tilt angle of the tilt photodetector. Fig. 1(b) The receiver horizontal photodetector center and each tilt photodetector center are equidistant from each other, and the tilt angle of each tilt photodetector is $\alpha$. $\beta_i$ is the angle between the projection of the line connecting the center of the horizontal photodetector and the center of the tilt photodetector on the *xoy* surface and the positive direction of the *x* axis.

### 2.2. Channel modeling

The indoor visible light positioning system selects LED lamps as the signal emission source, in which the radiation model introduces the Lambertian radiation model, equation (2) is the DC gain $H_{LOS}$ of the direct link channel:

$$H_{LOS} = \begin{cases} \dfrac{(m+1)A_{PD}}{2\pi d^2}\cos^m(\varphi)T_s(\psi)g(\psi)\cos(\psi), & 0 \le \psi \le \psi_{FOV} \\ 0, & \psi > \psi_{FOV} \end{cases} \tag{2}$$

where $A_{PD}$ is the effective receiving area of the PD, $m$ is the Lambertian emission order, $d$ is the distance from the LED to the PD, $\varphi$ is the emission angle of the LED, $T_s(\psi)$ is the optical filter gain, $g(\psi)$ is the non-imaging concentrator gain, $\psi$ and $\psi_{FOV}$ are the angle of incidence and field of view of the PD, respectively, $m$ and $g(\psi)$ can be calculated by equation (3):

$$\begin{cases} m = \dfrac{-\ln 2}{\ln\left(\cos\left(\varphi_{1/2}\right)\right)} \\ g(\psi) = \dfrac{n^2}{\sin\left(\psi_c\right)^2} \end{cases} \tag{3}$$

where $\varphi_{1/2}$ is the half-power half-angle of the LED, i.e., the radiant power of the LED light source at this angle is half of the center power, and $n$ is the internal refractive index of the concentrator.

In a reflective link considering irregular walls, the channel gain $H_{NLOS}$ can be calculated from equation (4):

$$H_{NLOS} = \begin{cases} \displaystyle\sum_i^N \frac{A_{PD}(m+1)\rho_i\Delta A}{2\pi^2 d_{1i}^2 d_{2i}^2}\cos^m(\varphi_{1i})\cos(\psi_{1i})\cos(\varphi_{2i})\cos(\psi_{2i})T_s(\psi_{2i})g(\psi_{2i}), 0 \le \psi_{2i} \le \psi_{FOV} \\ 0, \psi_{2i} > \psi_{FOV} \end{cases} \tag{4}$$

where $N$ is the number of all reflective walls divided by $\Delta A$, $\rho_i$ is the reflectivity of the wall, $d_{1i}$ is the distance from the LED to the wall reflective unit, $d_{2i}$ is the distance from the wall reflective unit to the PD, $\varphi_{1i}$ is the emission angle of the LED, $\psi_{1i}$ is the angle of incidence of the wall reflective unit, $\varphi_{2i}$ is the emission angle of the wall reflective unit, and $\psi_{2i}$ is the angle of incidence of the PD. The normal vector $\overrightarrow{n_{wi}}$ of the wall reflection unit can be expressed by equation (5):

$$\overrightarrow{n_{wi}} = (\cos(\alpha_{wi})\sin(\beta_{wi}), \sin(\alpha_{wi})\sin(\beta_{wi}), \cos(\beta_{wi})) \tag{5}$$

where $\alpha_{wi}$ and $\beta_{wi}$ are the azimuth and tilt angles of the wall reflector unit, respectively, then the corresponding cosine values of $\varphi_{1i}$, $\psi_{1i}$, $\varphi_{2i}$ and $\psi_{2i}$ can be expressed by equation (6):

$$\begin{cases} \cos(\varphi_{1i}) = h_{1i}/d_{2i} \\ \cos(\psi_{1i}) = \dfrac{(x_t - x_{wi})\cos(\alpha_{wi})\sin(\beta_{wi}) + (y_t - y_{wi})\sin(\alpha_{wi})\sin(\beta_{wi}) + (z_t - z_{wi})\cos(\beta_{wi})}{d_{2i}} \\ \cos(\varphi_{2i}) = \dfrac{(x_r - x_{wi})\cos(\alpha_{wi})\sin(\beta_{wi}) + (y_r - y_{wi})\sin(\alpha_{wi})\sin(\beta_{wi}) + (z_r - z_{wi})\cos(\beta_{wi})}{d_{2i}} \\ \cos(\psi_{2i}) = \dfrac{(x_{wi} - x_r)\cos(\alpha_r)\sin(\beta_r) + (y_{wi} - y_r)\sin(\alpha_r)\sin(\beta_r) + (z_{wi} - z_r)\cos(\beta_r)}{d_{2i}} \end{cases} \tag{6}$$

where $h_{1i}$ is the vertical height of the LED to the wall reflector unit and $(x_{wi}, y_{wi}, z_{wi})$ is the positional coordinates of the wall reflector unit.

In this paper, the primary reflections from the four walls in the room are considered, so the channel gain of the whole communication environment can be finally expressed by equation (7) as:

$$H_i = H_{LOS} + H_{NLOS} \tag{7}$$

In a visible light transmission link, the relationship between the received power $P_i$ of the receiver PD and the emitted power $P_t$ of the LED can be expressed by equation (8):

$$P_i = P_t H_i \tag{8}$$

## 3. Positioning principle

### 3.1. Deep residual shrinkage networks

Deep residual shrinkage networks incorporate three primary elements: deep residual networks, attention mechanisms, and soft threshold functions. In a standard neural network, as the number of layers increases, the gradient diminishes, leading to significant training challenges. To overcome this issue, deep residual networks utilize residual blocks. These blocks encompass multiple convolutional layers to learn feature representations, and they include a skip connection that enables the gradient to flow by adding the input directly to the output of the convolutional layers. This methodology enables the network to grasp incremental changes in features by computing the discrepancy between inputs and outputs. This mechanism facilitates the training of deep models, alleviating issues related to gradient vanishing and explosion. Refer to Fig. 2 for a visual representation of a basic residual shrinkage module.

The blue box houses an attention mechanism known as squeeze and

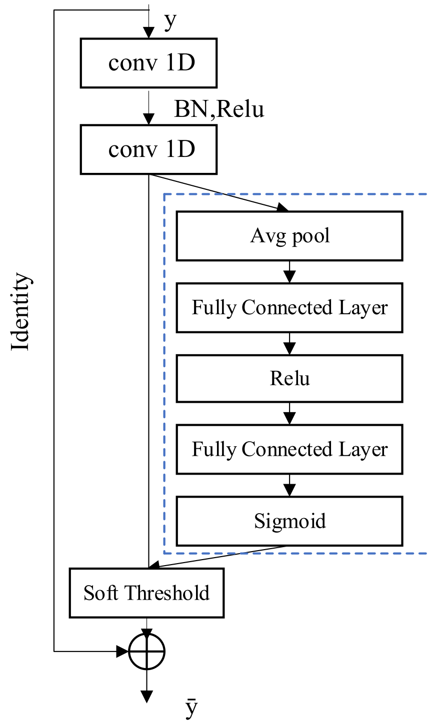



**Fig. 2.** Basic residual shrinkage module.

excitation (SE), designed to enhance the network's expressive capacity and overall performance by dynamically regulating the importance of information among channels. SE-Net is seamlessly combined with the residual module, providing an advantageous continuity of pathways across layers, which in turn simplifies the training of SE-Net. This integration allows SE-Net to acquire a set of thresholds essential for the soft-thresholding of individual feature channels to process the residuals effectively. The soft thresholding operation, as described, is implemented using a soft threshold function. This function is a nonlinear methodology frequently applied in signal processing, optimization algorithms, and sparse representations. Its mathematical definition is represented by equation (9).

$$soft\ threshold(x, \lambda) = sign(x)\max(|x| - \lambda, 0) \tag{9}$$

where $x$ is the input value and is a threshold parameter representing the sign of taking $x$. The function of the soft threshold function is to compare the absolute value of the input $x$ with the threshold value. When the absolute value is less than or equal to the threshold, the output is 0. When the absolute value is greater than $\lambda$, the output is either subtracted from or added to the original value, so that the part below the soft threshold has a certain smooth transition. Such a soft threshold operation preserves the information greater than the threshold and gradually reduces the information less than the threshold, resulting in a smoother output.

**Table 1**
Comparison of localization errors for different batch sizes.

| Batch size | MSE | $\bar{e}$/cm |
|---|---|---|
| 32 | 0.000201 | 1.76 |
| 64 | 0.000148 | 1.45 |
| 128 | 0.000115 | 1.02 |
| 256 | 0.000120 | 1.11 |

The comprehensive structure of the network model is illustrated in Fig. 3. Initially, one-dimensional optical power data undergoes input and processing through a convolutional layer with a $3 \times 1$ convolutional kernel, facilitating the extraction of pertinent features from the original dataset. Subsequently, the processed data are fed into the residual module to further enhance the extraction of effective features. The integration of the SE attention mechanism and soft thresholding within the residual module directs the network's focus towards crucial channels, thereby augmenting the localization accuracy of the model. Following these steps, connections are established with neurons in the fully connected layer after pooling and flattening layers. Ultimately, the corresponding coordinate values are output.

### 3.2. Performance indicators

The mean square error (MSE) and root mean square error (RMSE) metrics are employed to assess the performance and viability of the network model introduced in this study. MSE functions as the model's loss function, quantifying the deviation between the model's predicted values and the actual data. The utilization of MSE involves squaring the differences between predicted and actual values, placing emphasis on the squared discrepancies in outliers. This approach enhances the model's sensitivity to identify prediction errors. Equation (10) defines the calculation process for MSE.

$$MSE = \frac{1}{N}\sum_{k=1}^{N}\left[(\hat{x}_k - x_k)^2 + (\hat{y}_k - y_k)^2 + (\hat{z}_k - z_k)^2\right] \tag{10}$$

where $N$ is the total number of samples, $(x_k, y_k, z_k)$ is the true value of the $k^{th}$ sample point in the data set, and $(\hat{x}_k, \hat{y}_k, \hat{z}_k)$ is the predicted value of the $k^{th}$ sample point in the data set.

The root mean square error also evaluates the distinction between the model's predicted and actual values. It is determined by averaging the squares of the differences between the predicted and actual values and then computing their square root. An advantage of RMSE is its inherent feature to impose a greater penalty on larger error values, thereby averting the undue influence of significant errors on the overall fit. Equation (11) represents the computational process for RMSE.

$$e_i = \sqrt{(\hat{x}_i - x_i)^2 + (\hat{y}_i - y_i)^2 + (\hat{z}_i - z_i)^2} \tag{11}$$

where $(x_i, y_i, z_i)$ is the real coordinate of the j$^{th}$ reference point in the data set and $(\hat{x}_i, \hat{y}_i, \hat{z}_i)$ is the predicted coordinate of the $i^{th}$ reference point in the data set. Therefore, the average localization error $\bar{e}$ of the network model can be expressed by equation (12):

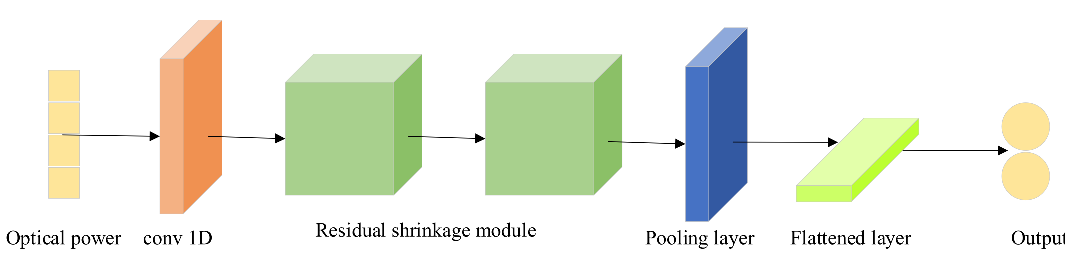



**Fig. 3.** Deep residual shrinkage network structure.

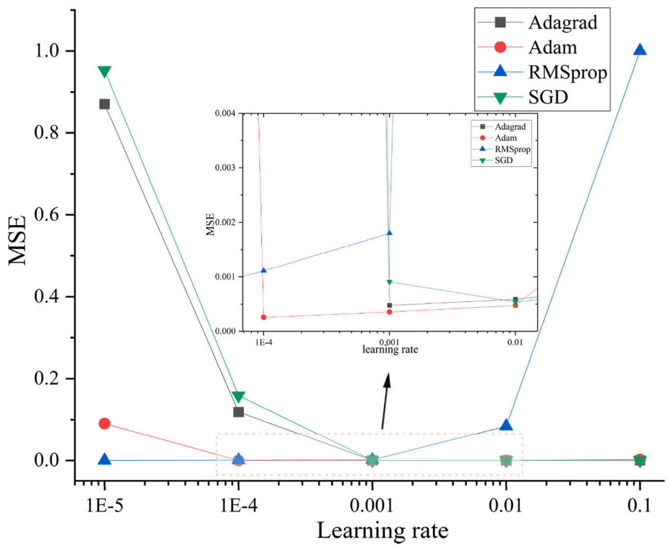



**Fig. 4.** Effect of each optimization algorithm on localization performance at different learning rates.

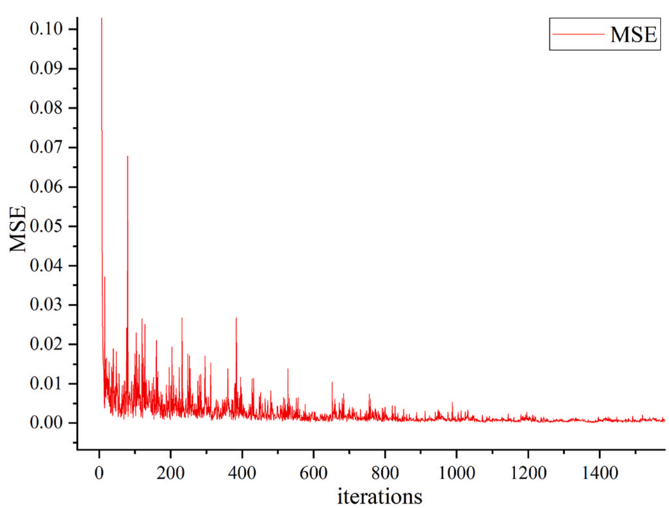



**Fig. 5.** Curve of loss function with the number of iterations.

$$\bar{e} = \frac{1}{N}\sum_{i=1}^{N} e_i \tag{12}$$

### 3.3. Network model parameter selection

Batch processing entails simultaneously processing multiple samples during each update of the model parameters, and the selection of batch size significantly influences the accuracy of the localization model. To conduct a comprehensive comparative analysis, we systematically examine batch sizes of 32, 64, 128, and 256, with specific results presented in Table 1. In cases where the batch size is too small, the gradient may exhibit instability, impeding network convergence and subsequently diminishing model accuracy. Conversely, setting the batch size too large compromises the network's generalization ability, resulting in less noticeable changes in model error. Therefore, the optimal performance of the network model in this study is achieved with a batch size set to 128.

The learning rate typically governs the magnitude of model weight updates during each iteration, playing a pivotal role in training deep learning models. Selecting an appropriate learning rate is crucial, and optimization algorithms are employed to regulate this rate. Hence, this study assesses the impact of different optimization algorithms on the localization performance of the network model by adjusting the learning rate, with the results presented in Fig. 4. Notably, the Adam optimization algorithm demonstrates the smallest mean squared error change when the learning rate is set to 0.0001. Based on this observation, we ultimately adopt the Adam algorithm with a learning rate of 0.0001 for training the network model.

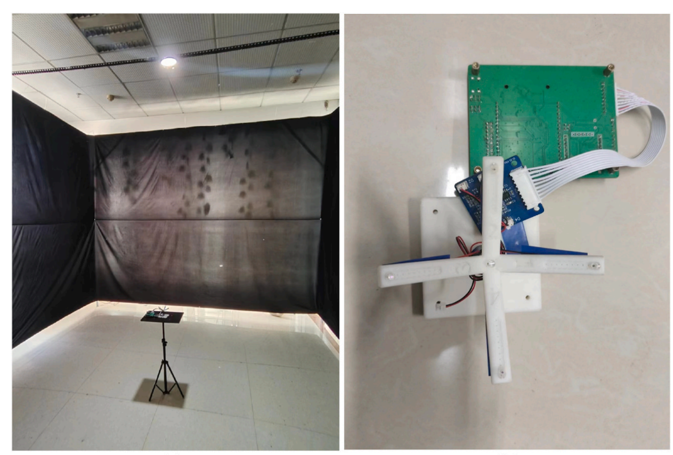
The term "number of iterations" refers to the repetitive execution of a set of operations or update steps, progressively aligning the network model with an optimal state. Factors influencing the setting of the number of iterations typically include network complexity, dataset size, and the inherent nature of the problem. While an increase in the number of iterations enhances result accuracy, it also escalates computational costs. In pursuit of optimal model parameters, this study compares mean

**Table 2**
Channel simulation parameters.

| Parameters | Values |
|---|---|
| Room size($m$) | 3.6 × 3.6 × 3 |
| LED position, ($m$) | (1.8,1.8,3) |
| Reflection coefficient, $\rho$ | 0.7 |
| Transmission power, $P$ ($W$) | 12 |
| Field of view angle, $\psi_c$ (°) | 90 |
| Effective accepted area, $A$ ($cm^2$) | 1 |
| Reflector cell size, $A_\omega$ ($m^2$) | 0.1 |
| Refractive index of lens, $n$ | 1.5 |
| Half power angle, $\varphi_{1/2}$ (°) | 60 |
| Wall tilt angle, $\alpha_\omega$ (°) | [0,180] |
| Wall rotation angle, $\beta_\omega$ (°) | [0,180] |
| PD Tilt angle, $\alpha$ (°) | 20 |
| Tilt PD placement angle,($\beta_2$, $\beta_3$, $\beta_4$)(°) | (0,90,180) |

**Table 3**
List of experimental equipment configuration.

| Parameters | Values |
|---|---|
| PD model | ALS-PDIC144-6C/L378 |
| PD Size (mm) | 3 |
| Number of PDs | 4 |
| Interval of PD (cm) | 5 |
| Wide supply voltage range ($V$) | 1.8–5.5 |
| LED Shape | orbicular |
| LED Size (mm) | 9 × 169 |
| Main control board | STM32F103C8T6 |
| Transmission module | ESP8266 |



(a) (b)

**Fig. 6.** Indoor localization experimental environment (a) experimental platform (b) receiver.

**Table 4**
Division spacing scheme.

| | Training data interval/cm | Prediction data interval/cm | $\overline{e}$ in simulation environment/cm | $\overline{e}$ in test environment/cm |
|---|---|---|---|---|
| Option I | 15 × 15 | 20 × 20 | 1.02 | 5.85 |
| Option II | 18 × 18 | 24 × 24 | 1.26 | 6.54 |
| Option III | 20 × 20 | 30 × 30 | 1.36 | 7.58 |

squared error values under various iteration numbers, maintaining consistency in other parameters. The loss function curve with respect to the number of iterations is depicted in Fig. 5. Notably, the MSE fluctuation remains relatively stable beyond 1300 network iterations, as evident in the figure. Consequently, this study sets the maximum number of iterations for the network model at 1300.

## 4. Simulation, experimentation and analysis

### 4.1. Simulation and experimental environment

To substantiate the viability of the algorithm introduced in this paper, we established both simulation and experimental environments following the configuration depicted in Fig. 1. Specific parameters for both simulation and experiments are detailed in Tables 2 and 3. Within the designated sample room, a solitary white LED was installed directly above, with a receiver strategically positioned on a tripod. This setup facilitated easy height adjustments for the receiver, enabling the capture of optical power values and the construction of a precise fingerprint library. The layout and comprehensive details of the indoor localization experimental environment are illustrated in Fig. 6.

Fig. 6 (b) illustrates the receiver employed in this investigation, comprising multiple photodetectors and a central control module. The photodetectors receive an optical signal from a designated point, convert it into an electrical signal, and amplify the signal to the requisite level using an amplification circuit. This amplification serves the purpose of ensuring a stable and suitable input for a subsequent low-pass filter, effectively eliminating potential noise interference. Ultimately, the electrical signal undergoes conversion to the corresponding optical

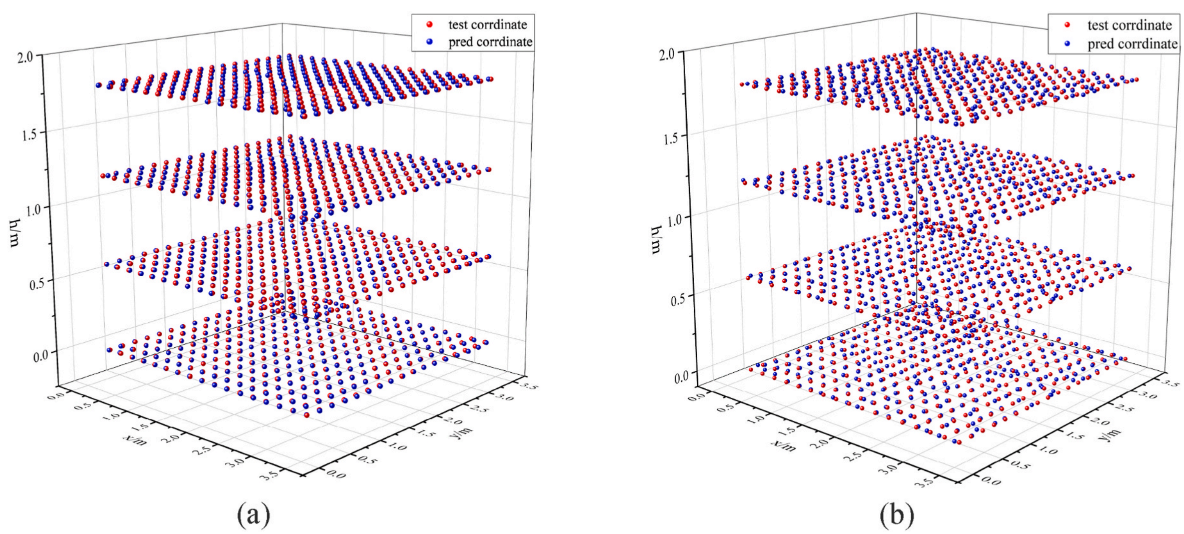



**Fig. 7.** Distribution of real and predicted coordinates at different heights (a) simulation phase and (b) experimental phase.

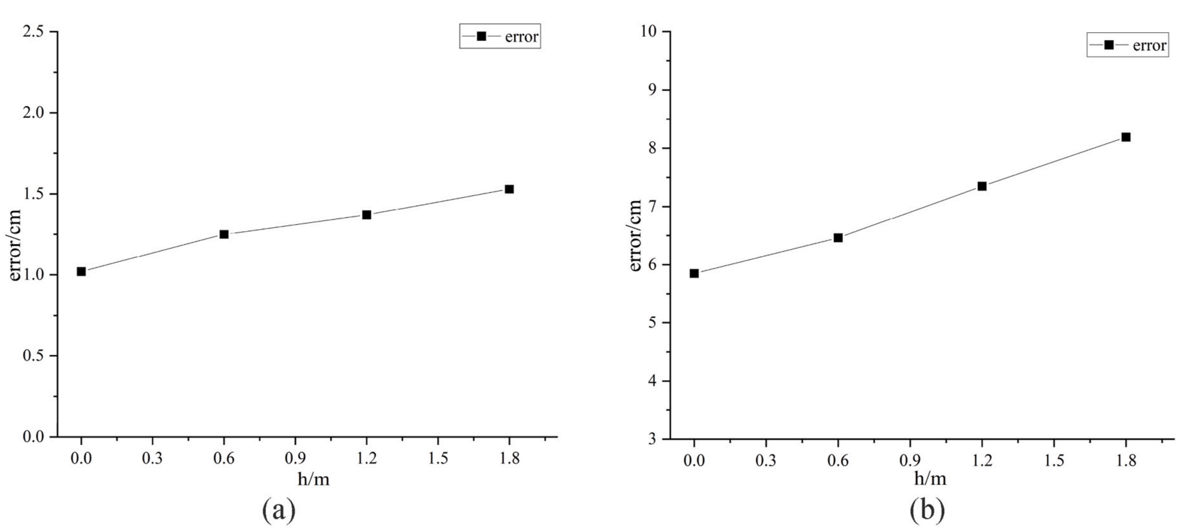



**Fig. 8.** Error variations at different heights of the receiving end from the ground (a) simulation phase and (b) experimental phase.

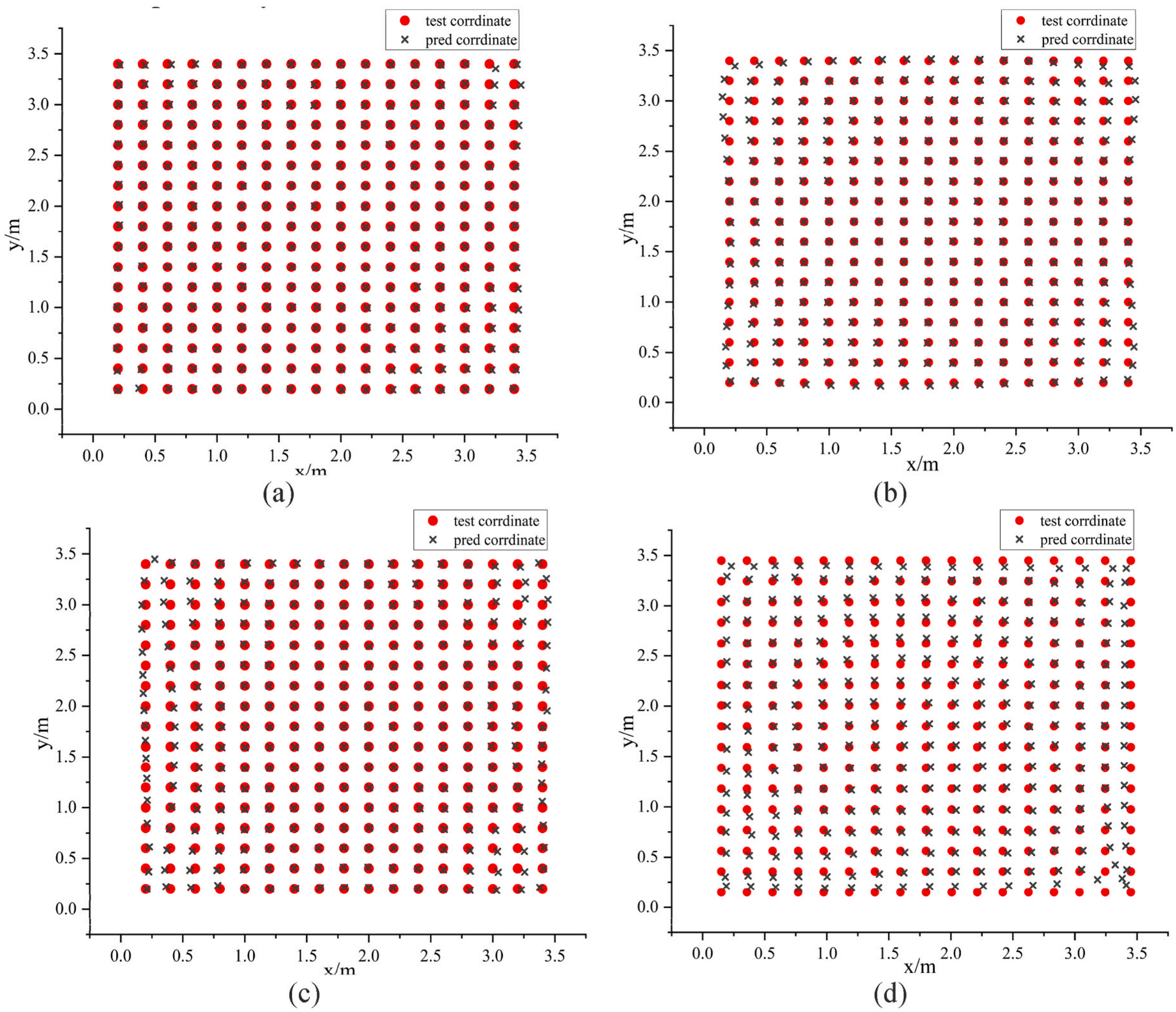



**Fig. 9.** Error distribution of different localization methods in simulation phase (a) DRSN (b) LSTM (c) GRU (d) CNN.

power value through an analog to digital converter (ADC) before being transmitted back to the computer for storage, essential for subsequent model training.

### 4.2. Effect of fingerprint spacing on localization error

The generation of fingerprints through an offline database plays a pivotal role in the localization process using the deep residual shrinkage network. The localization method involves two distinct phases: offline and online. Initially, during the offline phase, the room is segmented into defined intervals for gathering light intensity signals, which are stored in the database. Following this, a predictive model is trained to estimate the coordinate values for the VLP test points. Subsequently, in the online phase, new test coordinate points, distinct from the previously trained ones, are incorporated into the trained prediction model. The proposed algorithm's performance is evaluated by comparing the predicted values to the actual values. This section presents three distinct division interval schemes, with specific data detailed in Table 4.

Table 4 presents a clear trend where the average localization error escalates with the increase in fingerprint spacing, observed in both simulation and experimental phases. This rise in error is attributed to the optical power values undergoing various reflections and refractions, creating multiple paths to reach the receiver. Consequently, signals traveling through different paths generate phase differences at the receiver. As a result, a wider spacing impedes the localization system from capturing subtle variations and features, thereby contributing to an increase in localization error.

### 4.3. Effect of receiver height on localization error

To assess the algorithm's generalization capabilities, we varied the receiver's height (h) to 0 cm, 60 cm, 120 cm, and 180 cm. As depicted in Fig. 7, we obtained distributions of true and predicted coordinates for different heights through the training and prediction phases of the localization model.

Fig. 8 illustrates the error variation at the receiving end for different heights from the ground. It is evident that a reduced distance between the receiving and transmitting ends corresponds to a greater error between the actual and predicted coordinate values. This phenomenon arises due to the decreasing distance from the receiving end to the LED light source, causing an increase in the emission angle of the LED light source. Consequently, the values of the DC gains $H_{LOS}$ and $H_{NLOS}$ diminish, resulting in a reduced intensity of the received light signal at the receiving end and, consequently, an increased localization error.

### 4.4. Comparison of different algorithms

In this segment, the efficacy of the proposed method is further substantiated through a comparison and analysis between the deep residual shrinkage network (DRSN) and prediction models including the convolutional neural network (CNN), long short-term memory network (LSTM), and gated recurrent unit network (GRU). Setting the receiver

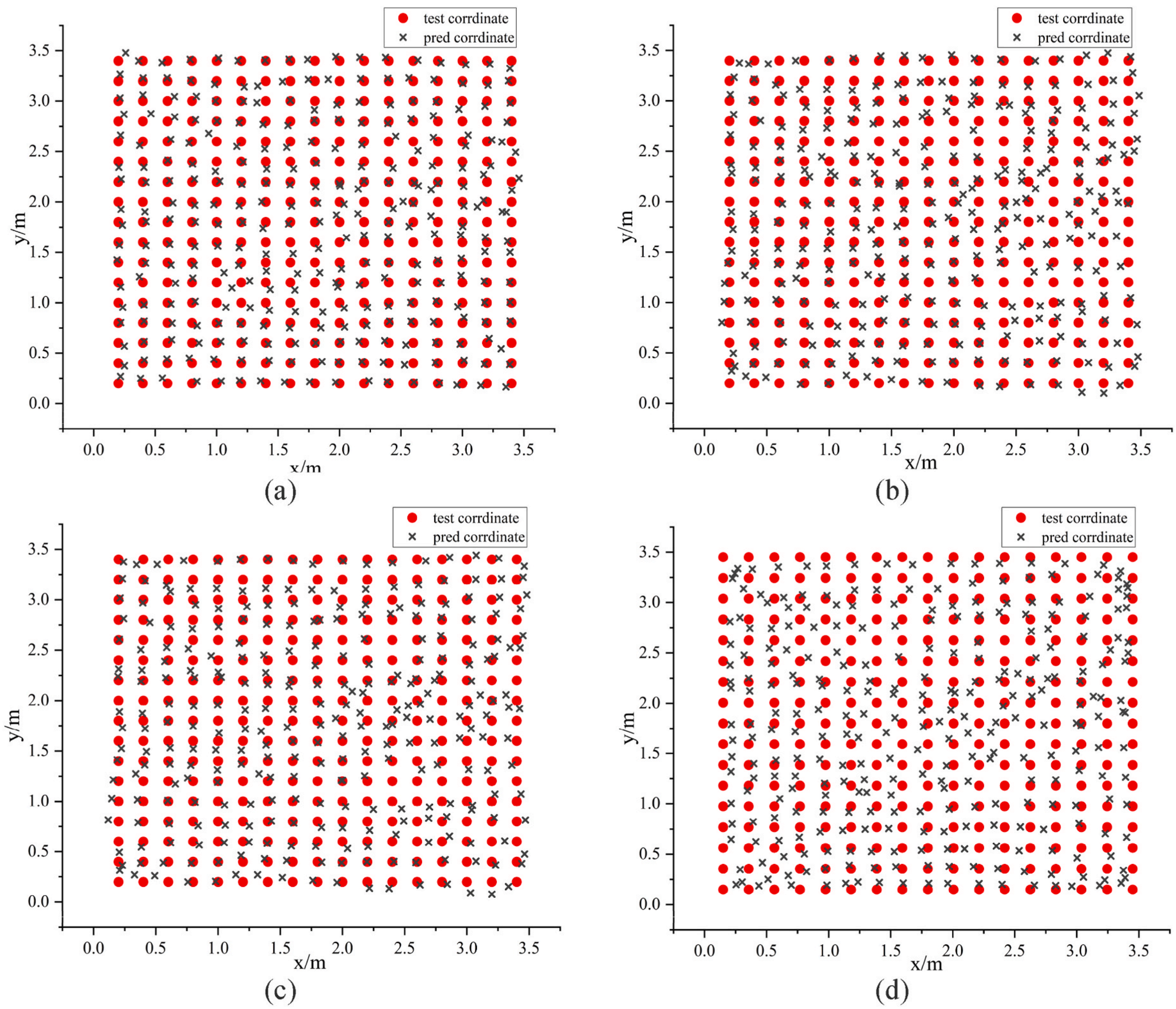



**Fig. 10.** Error distribution of different localization methods in experimental phase (a) DRSN (b) LSTM (c) GRU (d) CNN.

**Table 5**
Comparison of errors of different algorithms.

| Positioning algorithm | Simulation phase | | Experimental phase | | Prediction time/s |
|---|---|---|---|---|---|
| | $\bar{e}$/cm | Max. positioning error/cm | $\bar{e}$/cm | Max. positioning error/cm | |
| DRSN | 1.02 | 4.35 | 5.85 | 26.41 | 0.0133 |
| LSTM | 2.04 | 9.29 | 7.83 | 41.82 | 0.0147 |
| GRU | 2.11 | 9.32 | 7.91 | 45.61 | 0.0142 |
| CNN | 4.14 | 19.41 | 11.92 | 35.22 | 0.0176 |

height as h = 0 m and adopting Option I for division intervals, the error distribution plots of distinct localization methods in both simulation and experimental stages are presented in Figs. 9 and 10, respectively. Detailed data are available in Table 5.

Table 5 demonstrates that, in both simulation and experimental phases, the deep residual shrinkage network exhibits a substantial advantage in terms of localization accuracy compared to the other three algorithms. Despite not experiencing a significant increase in localization prediction time compared to the alternatives, DRSN efficiently accomplishes the localization task within a millisecond time scale, maintaining real-time performance akin to the other algorithms. Additionally, Figs. 9 and 10 reveal outliers in the localization results. This may stem from interference during dataset collection, such as dark current or external factors affecting the receiver, leading to sudden fluctuations in the received optical power at specific points and resulting in suboptimal training and prediction outcomes. Notably, the deep residual shrinkage network effectively mitigates model sensitivity to noise through the use of a soft threshold function, proving advantageous in handling noise and uncertainties in actual data. Furthermore, employing residual learning allows the network to acquire a deeper feature representation, and the integration of the squeeze and excitation attention mechanism enhances the network's focus on essential feature channels. A comparison of their maximum localization errors highlights the superiority of the DRSN algorithm over the other three algorithms for handling outliers.

To facilitate a clearer comparison of positioning accuracy among different localization algorithms, Fig. 11 illustrates the cumulative probability distribution of localization errors in both simulation and experimental phases. In the simulation phase, the proposed method in this paper exhibits exceptional performance, with 90% of localization errors confined within a mere 2.35 cm and remaining within the 5 cm threshold. In contrast, the LSTM, GRU, and CNN-based localization methods report 90% localization errors at 3.76 cm, 3.81 cm, and 6.42 cm, respectively. Moving to the experimental phase, the DRSN algorithm achieves a 90% error below 10 cm, while the LSTM, GRU, and CNN-based methods, under identical conditions, register 90% localization errors within 15.63 cm, 16.01 cm, and 21.62 cm, respectively. Hence, for the single LED localization system in this study, the DRSN algorithm outperforms its counterparts in terms of accuracy.

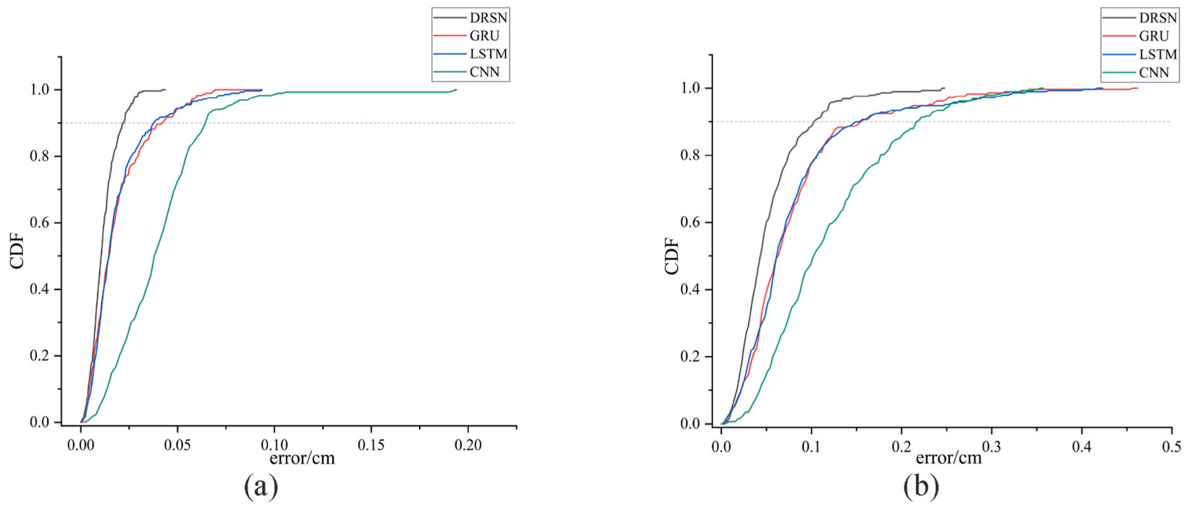



**Fig. 11.** Comparison of CDF for different localization methods (a) Simulation phase (b) Experimental phase.

## 5. Conclusion

Addressing the challenge of neural networks inadequately addressing the robustness issues in received signal strength, we propose an indoor visible light positioning algorithm based on the deep residual shrinkage network. The algorithm's localization accuracy is comprehensively analyzed across various datasets and receiver heights. The examination reveals a gradual reduction in the error between true and test coordinates as the data division interval and receiver height decrease. Additionally, a comparative assessment against LSTM, GRU, and CNN algorithms demonstrates that DRSN surpasses these counterparts not only in terms of localization accuracy but also in maintaining a comparable level of real-time performance. The integrated simulation and experimental outcomes reveal an average localization error of 1.02 cm and a maximum error of 4.35 cm in a 3.6 m × 3.6 m × 3 m space during the simulation phase. In the experimental phase, the average localization error is 5.85 cm, with 90% of errors below 10 cm, and a localization prediction time of 0.0133 s. These findings underscore DRSN's competitiveness in accuracy while meeting real-time demands, offering a viable and reliable solution for practical applications in indoor positioning systems.

## CRediT authorship contribution statement

**Shuai Li:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Ling Qin:** Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Desheng Zhao:** Supervision, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Xiaoli Hu:** Supervision, Methodology, Investigation, Formal analysis, Data curation.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ling qin reports financial support was provided by the National Natural Science Foundation of China and the Inner Mongolia Autonomous Region Applied Technology Research and Development Fund. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The data that has been used is confidential.

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